

Model Report: Multi-Channel CNN Patch-Level

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Script: /home/mmann1123/Documents/github/South_Africa_Crop_Comp/deep_learn/src/Deep Learning/Patch Level/Multi-Channel CNN Patch-Level

Data Split:

train_count: 21741
test_count: 830
random_seed: 42
split_method: field-based (patch-level)

Metrics (test set):

Accuracy: 0.7072
Cohen Kappa: 0.5927
F1 Weighted: 0.7007
F1 Macro: 0.6284

Hyperparameters:

target_size: [128, 128]
batch_size: 64
epochs: 20
optimizer: Adam
loss: sparse_categorical_crossentropy
channels: 60

Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Barley	0.515	0.696	0.592	125
Canola	0.930	0.742	0.825	89
Lucerne/Medics	0.859	0.910	0.884	356
Small grain grazing	0.379	0.259	0.308	85
Wheat	0.568	0.503	0.533	175

Confusion Matrix

